NEXUS · Smart Gates & Terminals · Port Scene Analytics (PPS 3.6)

SMART GATE

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ABSTRACT

Port gates are inspected by hand today - plate, container code, seal, corrosion - one truck at a time, with no shared record linking the checks. Smart Gate integrates four independent AI models developed within NEXUS into one platform that triggers all four inspections together and merges their results into a single Inspection Event, tied to the same truck, gate, and moment.

1. CONTEXT

NEXUS is a national innovation agenda funded by Portugal's Recovery and Resilience Plan, led by the Port of Sines Authority with 35 partners. Smart Gate belongs to its Smart Gates & Terminals pillar.

2. CONTRIBUTION

Smart Gate is the platform that correlates all four readings into a single inspection event per gate passage, a correlation no individual module can make on its own.

3. THE INTEGRATION CHALLENGE

These four technologies don't share a language: different response times, different formats, some instant, some take seconds. Smart Gate captures the right camera feed for each one, runs them in parallel, and proves all four results belong to the exact same truck, at the exact same gate, at the exact same moment.

4. THE FOUR TECHONOLOGIES

MOD. 1 - OCR [1]

Container code

Reads the ISO 6346 code.



MOD. 2 - CDD [2]

Corrosion

Estimates corrosion from the image.



MOD. 3 - LPR [3]

License plate

Reads the truck's plate automatically.



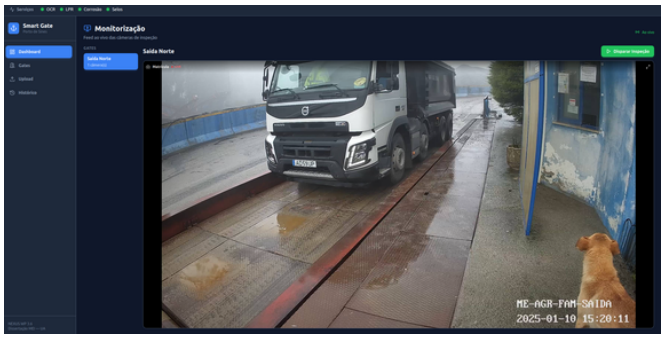
MOD. 4 - SEAL

Seal detection

Checks seal presence.



5. SOLUTION



Real-time camera stream, as seen by the operator before an inspection fires.

6. REFERENCES

[1] J. Santos, D. Canedo, A. J. R. Neves, "Automating Code Recognition for Cargo Containers," Electronics, vol. 14, no. 22, art. 4437, 2025. doi.org/10.3390/electronics14224437

[2] D. Ornelas, D. Canedo, A. J. R. Neves, "Semantic Segmentation of Corrosion in Cargo Containers Using Deep Learning," Sustainability, vol. 17, no. 14, art. 6480, 2025. doi.org/10.3390/su17146480

[3] D. Ornelas, J. Santos, D. Canedo, A. J. R. Neves, "License Plate Recognition in Port Scene Analytics," RECPAD 2025, Aveiro, Portugal, pp. 159–160.

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